AirSpy HF+ and HF+ Discovery

A Technical Analysis of a High-Performance Software-Defined HF/VHF Receiver

Architecture · Signal Chain Design · Dynamic Range
Engineering Trade-offs · Performance Analysis

KEY PERFORMANCE HIGHLIGHTS

Blocking Dynamic Range (HF):	110 dB — best-in-class consumer SDR
Minimum Detectable Signal (HF):	" 140 dBm (500 Hz noise bandwidth)
System Noise Figure:	~0.75 dB (HF+ Discovery optimised)
Alias/Image Rejection:	108 dBc
IIP3 (max gain, HF):	+15 dBm
ADC Resolution (analog):	24-bit equivalent (sigma-delta)
Decimated Output:	18-bit effective !’ 2×16-bit IQ USB
Output Sample Rate:	768 kSps / 600 kHz alias-free BW
Harmonic Rejection:	Up to 21st harmonic (PHR mixer)
Frequency Coverage:	9 kHz – 31 MHz (HF), 60 – 260 MHz (VHF)

Abstract

The AirSpy HF+ and its successor, the HF+ Discovery, represent a significant advance in low-cost, high-performance software-defined radio (SDR) receiver design for the HF and lower VHF spectrum. By combining a selectable bandpass preselector filter bank, a multi-stage automatic gain control (AGC) architecture employing both active variable gain amplification and passive attenuation, a low-noise amplifier (LNA) delivering approximately 20 dB of gain, a polyphase harmonic rejection (PHR) mixer operating at a low intermediate frequency (IF) to eliminate DC spur problems, a fourth-order multi-bit sigma-delta (:2 b• analogue-to-digital converter providing up to 24-bit equivalent analogue resolution decimated to 18-bit output, and a cascaded CIC/FIR decimation chain delivering 2×16-bit IQ output at 768 kSps, these receivers achieve a blocking dynamic range of 110 dB on HF — a figure that was unprecedented for consumer-grade SDR hardware at the time of their release.

This paper examines each stage of the signal chain in technical depth, discusses the engineering trade-offs made, identifies potential improvements, and situates the design within the broader context of high-performance HF receiver architecture. Where internal design details are not publicly disclosed, clearly-labelled engineering hypotheses are provided, consistent with published measurement data and known component specifications.

1. Introduction

Software-defined radio (SDR) has democratised spectrum monitoring, amateur radio experimentation, and signals intelligence research. However, for many years the dominant low-cost SDR platform — the RTL-SDR dongle based on the Rafael Micro R820T2 chip — suffered from a blocking dynamic range of only 50–60 dB on HF. This rendered it largely unsuitable for serious shortwave monitoring in the presence of strong AM broadcast stations, which routinely exceed 100 dB;ÅböÒ B „b –â ÷ VÆ FVB &Vv–öç0.

AirSpy, in collaboration with STMicroelectronics, addressed this limitation with the HF+ (released 2017) and the HF+ Discovery (released 2019). The key insight driving the design was that the core performance bottleneck was not digital processing capability but the analogue front-end: noise figure, linearity, and the quality of the analogue-to-digital conversion.

By leveraging the STMicroelectronics STA709 — an automotive-grade digital tuner IC featuring dual high-dynamic-range sigma-delta ADCs and an integrated polyphase harmonic rejection mixer — the AirSpy team achieved a Minimum Detectable Signal (MDS) of " 140 dBm and a blocking dynamic range of 110 dB on HF frequencies, both of which were unprecedented in the consumer SDR market.

The automotive heritage of the STA709 is highly relevant to this application. Automotive AM/FM tuners must simultaneously reject multi-kilowatt MW broadcast transmitters while recovering weak DX signals — a dynamic range challenge directly analogous to HF monitoring. This application context drove STMicroelectronics to implement exceptional noise-shaping ADC architectures in the STA709 that could be directly exploited by the AirSpy design team.

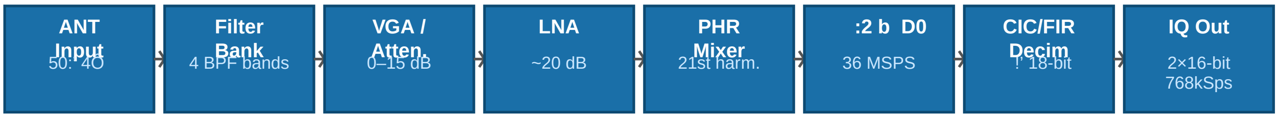

Figure: Simplified signal cha

2. Input Preselector Filter Bank

The first element in the signal chain after the antenna connector is the preselector filter bank. This is the most consequential stage for overall receiver performance in a dense RF environment, as it determines both the noise figure floor and the receiver's immunity to strong out-of-band signals.

2.1 Purpose and Topology

A preselector serves two related but distinct purposes. First, it limits the total RF power delivered to subsequent stages, protecting them from strong out-of-band signals that could cause gain compression or intermodulation. Second, it reduces the noise bandwidth presented to the LNA, preventing aliasing of wideband noise into the IF passband.

In both the HF+ and Discovery, the preselector consists of four selectable bandpass filters (BPFs) covering the HF range from 9 kHz to 31 MHz, plus a separate path for the VHF range (60–260 MHz). The filter bank is software-switched based on the currently tuned frequency.

Note:
The exact filter topology (lumped LC, transmission-line, ceramic resonator) has not been publicly disclosed by Airspy. Based on the frequency range, compact form factor, and steep skirts observed in published VNA measurements by independent reviewers, a multi-element LC bandpass design with four to six poles per band is the most probable implementation.

2.2 Filter Band Assignments

Band	Frequency Range	Primary Use	Key Benefit
1	9 kHz – 5 MHz	LF/MF, low HF	Rejects AM broadcast into LF
2	5 – 10 MHz	40m/60m amateur, intl. broadcast	Rejects high-power above 10 MHz
3	10 – 17 MHz	20m/17m amateur, shortwave	Rejects tropical shortwave IM
4	17 – 31 MHz	Upper HF, 10m–15m amateur	Rejects MW/LF intermodulation
VHF	60 – 260 MHz	FM broadcast, aviation, weather sat	Separate signal path

2.3 Insertion Loss and Noise Figure Impact

The preselector filters in the HF+ Discovery are reported to have approximately 1.4 dB of insertion loss. By Friis' cascade noise formula, this loss directly contributes to the system noise figure. However, with the LNA providing 20 dB of subsequent gain, this contribution is heavily diluted at all later stages.

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Friis cascade noise figure formula:
  F_total = F1 + (F2 - 1)/G1 + (F3 - 1)/(G1·G2) + ...

With filter insertion loss = 1.4 dB  (linear noise factor F1 = 1.38)
     LNA noise figure    = 0.75 dB  (linear noise factor F2 = 1.19)
     LNA gain            = 20 dB    (linear gain G1 = 100):

  F_total "H 1.38 + (1.19 - 1)/100 "H 1.38 + 0.002 "H 1.382
  NF_total "H 10·log • €(1.382) "H 1.4 dB

The LNA contribution to system NF is only 0.002 in noise factor terms.
The preselector's insertion loss dominates the cascade — not the LNA.
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This reveals a fundamental design principle: the filter's loss dominates the overall system noise figure, not the LNA. This is why the HF+ Discovery — which integrates the preselector with 1.4 dB insertion loss directly on-board —

shows a slightly higher noise figure than the original HF+ used without a preselector, but the tradeoff in improved dynamic range and selectivity is highly favourable in real-world operating environments.

Potential Improvement:
More aggressive bandpass filters (7- or 9-pole designs) could improve adjacent-band rejection by 15–25 dB. However, each additional filter pole adds approximately 0.1–0.3 dB insertion loss, which directly increases system NF. A hypothetical 7-pole filter adding 0.5 dB extra loss would increase system NF from 1.4 dB to 1.9 dB — a measurable sensitivity degradation for a marginal improvement in adjacent-channel rejection.

3. Variable Gain Amplifier (VGA) and Attenuation Stage

Following the preselector, the signal enters the variable gain / attenuation stage. This stage is central to the HF+'s ability to handle signals spanning many orders of magnitude in power without either clipping the ADC on strong signals or burying weak signals in the noise floor.

3.1 Function of the VGA

The VGA serves a dual role: it can provide gain (amplifying weak signals before the ADC) or act as a controlled attenuator (reducing strong signals to prevent ADC overload). In the HF+ architecture, gain control is applied in discrete steps — nominally 0 to 15 dB of control range with approximately 1 dB resolution — software-commanded by the DSP-based AGC system running within the STA709.

When the AGC control value is at its minimum, the VGA is applying maximum attenuation to protect the ADC from strong signals. As the control value increases, attenuation is reduced and then gain is applied to improve sensitivity for weak signals. The transition through unity gain occurs at the mid-point of the control range.

3.2 The Noise Figure Trade-off of Active VGA

Using an active VGA as an attenuator introduces a fundamental noise figure penalty not present with a purely passive attenuator. A passive resistive or switched attenuator has a noise figure exactly equal to its insertion loss — it adds no excess noise beyond the thermally unavoidable Johnson-Nyquist noise of its resistive elements. An active VGA in attenuation mode still contributes the noise of its active devices on top of the attenuation function.

However, the AirSpy design places the VGA after the LNA in the signal chain. By the time the signal reaches the VGA, the cascade noise figure is dominated by the early stages. The incremental noise added by a VGA positioned after a 20 dB LNA stage is attenuated by that gain when referred to the system input:

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If VGA in attenuation mode contributes excess NF of ~3 dB beyond attenuation value,
and the preceding LNA has 20 dB gain (G1 = 100):

VGA contribution to input-referred system NF = (F_VGA - 1) / G1

At maximum attenuation (15 dB) with active device NF (F_VGA "H 2 for typical CMOS):
  VGA contribution "H (2 - 1) / 100 = 0.01  !’ 0.04 dB additional system NF

This is negligible in practice. The dominant NF contribution remains
the preselector insertion loss (1.4 dB) in the cascade.
```

3.3 Why Not a Pure Passive Attenuator?

A purely passive attenuator (switched resistive pads, PIN-diode switched L-pads, or relay-switched pads) placed before the LNA would offer superior noise performance in high-signal conditions, since it introduces no excess active device noise. However, such a design carries significant practical disadvantages:

- Impedance matching: Each switched attenuation pad must maintain 50:’ Ö F6, 7&÷72 F†R gVÆÂ g&W VVæ7’ ange (9 kHz to 260 MHz); achieving this with fine-grained steps requires careful design and potentially multiple switching elements per step.
- Noise figure degradation: Any passive loss before the LNA adds directly to system NF (not divided by preceding gain). A 10 dB passive attenuator placed before the LNA would add exactly 10 dB to system NF, destroying sensitivity.
- Switching artefacts: PIN diode or relay switches introduce transient glitches during gain switching that can cause momentary saturation of downstream stages.
- PCB area and cost: A full switched attenuator bank with adequate range requires additional components and board space.

The active VGA approach, positioned after the LNA, provides the required dynamic range management with a compact circuit that adds negligible noise in the overall cascade — a well-reasoned engineering compromise.

Potential Improvement:

A parallel passive attenuation path, engaged only for extremely strong signals by bypassing the LNA entirely (e.g., a 20 dB fixed passive pad switched in when input power exceeds " 20 dBm), would theoretically eliminate all active noise contribution during strong-signal conditions. This approach is used in some professional HF receivers. The added relay/switch complexity, switching time penalty, and the fact that the VGA contribution is already <0.05 dB make it a minor improvement for specialist use cases only.

The HF+ implements three interlocked AGC loops that cooperate to maintain optimal performance across all signal conditions:

1. Analog IF AGC: Controls the VGA gain immediately before the ADC input. This is the primary loop for preventing ADC saturation. Threshold setting allows the AGC to tolerate +3 dB extra signal before stepping to the next attenuation level, improving stability on strong signals.
2. Digital IF AGC: Operates in the digital domain after the ADC, scaling the data stream. Activates when the digitised signal exceeds " 6 dBFS, providing additional headroom management without modifying the analogue gain.
3. Narrowband AGC: Acts on the final filtered signal within the selected VFO (receiver) bandwidth. Optimises gain distribution for the specific channel of interest rather than the entire IF passband.

   This cascaded arrangement allows the receiver to simultaneously optimise for weak-signal sensitivity (by maximising gain when no strong signals are present) and strong-signal linearity (by rapidly backing off gain when large signals appear), without requiring any manual control adjustment by the user.

4. Low Noise Amplifier (LNA)

The LNA is the most noise-critical active stage in the receive chain. According to Friis' cascade noise formula, its noise figure and gain set the fundamental sensitivity limit of the receiver — since it is the first gain element after the passive (noise-adding) preselector.

4.1 Component: MiniCircuits PSA4-5043+

Independent technical analysis of the AirSpy HF+ hardware has identified the LNA as a MiniCircuits PSA4-5043+ monolithic microwave integrated circuit (MMIC). This device is a high-gain, low-noise pseudomorphic high-electron-mobility transistor (pHEMT) amplifier, originally specified for 50 MHz to 4 GHz operation.

Parameter	Specification	Conditions
Frequency range	50 MHz – 4 GHz	Rated range
Gain at 500 MHz	~20–21 dB	Vcc = 3 V
Noise Figure at 500 MHz	~0.75 dB	Vcc = 3 V
OIP3 at 1 GHz	+27 dBm	Vcc = 3 V
1 dB compression (output)	+13 dBm	Vcc = 3 V
Supply current	~40 mA	Vcc = 3 V
Package	SOT-86 / SC-70	3-terminal MMIC

Note:
The PSA4-5043+ is specified from 50 MHz. For HF operation (9 kHz – 31 MHz), the device is used below its rated frequency floor. At HF frequencies, MMIC gain typically increases slightly (common-emitter/common-source gain rises as frequency decreases toward DC), and the noise figure may differ from the VHF specification. The measured HF+ system NF of ~0.75 dB at HF suggests the pHEMT device performs excellently below its rated frequency range.

4.2 Gain and Noise Figure in the HF Chain

The LNA provides approximately 20–21 dB of gain. This large gain has two critical consequences for system performance:

First, regarding noise figure: By Friis' formula, 20 dB of gain means that the noise contribution of every subsequent stage (VGA, mixer, ADC) is divided by a factor of 100 before being referred to the antenna input. This effectively makes the system noise figure equal to the cascade of preselector loss and LNA noise figure alone — all subsequent stages are rendered insignificant. Measured system NF of approximately 0.75–1.4 dB (depending on configuration) is consistent with this analysis.

Second, regarding dynamic range: High LNA gain means that the ADC input sees a stronger version of the RF signal. This is beneficial for weak signals (moves them above the ADC quantisation noise floor) but means the system reaches ADC saturation at lower antenna input powers. The multi-stage AGC manages this tension by backing off LNA gain when strong signals are present. The IIP3 of +15 dBm (measured at the HF+ system input at maximum gain) reflects the excellent linearity of the pHEMT device and demonstrates the effectiveness of the gain-chain optimisation.

5. Polyphase Harmonic Rejection Mixer and IF Architecture

After the LNA, the signal enters the mixer stage, which translates it from RF to a low intermediate frequency (IF). The choice of mixer topology and IF strategy are critical design decisions that profoundly affect receiver performance, particularly the ability to handle the HF spectrum without a frequency-tracking front-end filter.

5.1 The DC Spur Problem in Zero-IF Receivers

Many low-cost SDRs use a direct-conversion (zero-IF) architecture, where the local oscillator (LO) is set exactly to the received frequency, placing the desired signal at DC (0 Hz) in the baseband. This architecture is attractive for its simplicity and lack of image rejection requirements, but suffers from two severe fundamental problems:

1. LO self-mixing / DC offset: Inevitable leakage of the LO signal into the RF port — or direct pickup by the antenna — causes the LO to mix with itself, producing a large DC offset at the mixer output that can saturate downstream stages and permanently obscure signals near the tuned frequency.
2. 1/f (flicker) noise: Active devices in the mixer contribute flicker noise that rises steeply as frequency approaches DC, creating a noise mountain at exactly the frequency where the wanted signal is placed in a zero-IF receiver.

5.2 The HF+ Solution: Low-IF Architecture

The HF+ avoids zero-IF entirely by using a low-IF architecture: the LO is offset from the desired signal frequency by a small but non-zero IF. The wanted signal appears at this offset frequency rather than at DC, safely away from both the DC offset and the 1/f noise peak. The IF is then removed in the digital domain, where 1/f noise does not exist.

Hypothesis / Engineering Inference:
The specific IF value used in the HF+ is not publicly disclosed by Airspy. Based on the ADC sampling rate of 36 MSPS, the published alias-free bandwidth of 600 kHz, and typical low-IF design practice for this class of sigma-delta converter, the IF is estimated to be in the range of 200 kHz to 2 MHz. The firmware adaptively shifts the IF to avoid correlation with sigma-delta modulator clock harmonics and sampling-rate spurs — a key technique for maintaining a clean noise floor across the entire operating range.

5.3 Polyphase Harmonic Rejection (PHR) Mixer

Standard mixers multiply the RF signal with the LO to produce the desired IF, but they also produce mixing products at LO harmonics (2×LO, 3×LO, etc.). Signals at these harmonic frequencies alias down to the same IF as the desired signal, creating spurious responses. Conventionally, this is suppressed by front-end tracking bandpass filters, but such filters add cost, insertion loss, and architectural complexity.

The STA709's PHR mixer uses a polyphase technique to achieve harmonic rejection without tracking filters. Multiple phase-shifted versions of the LO are combined with appropriate weightings so that signals at specific harmonic frequencies cancel at the output, while the fundamental mixing product adds coherently.

For the implementation in the STA709, harmonic rejection is achieved up to the 21st harmonic. This means that any signal at an odd harmonic of the LO frequency up to 21×f_LO is rejected from the IF output. For HF use (up to 31 MHz), even the 3rd harmonic response falls at 93 MHz, well outside the HF band where it is also heavily attenuated by the HF preselector filter. The PHR mixer thus completely eliminates the need for a tracking bandpass filter, maintaining a low-loss, broadband signal path to the ADC.

```
For a signal tuned to f_RF = 10 MHz (LO at ~10 MHz plus small IF offset):
  3rd harmonic response: at f_in = 30 MHz  (attenuated by HF preselector)
  5th harmonic response: at f_in = 50 MHz  (VHF — strongly rejected by HF BPF)
 21st harmonic response: at f_in = 210 MHz (rejected by PHR + preselector)

For f_RF = 28 MHz (top of HF amateur band):
  3rd harmonic: at f_in = 84 MHz  (FM broadcast band — rejected by HF preselector)

Result: No tracking filter is required across the entire HF coverage range.
The HF preselector's band-limited response handles the few cases where a
harmonic response could fall in-band for the highest tuning frequencies.
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6. The Sigma-Delta Analogue-to-Digital Converter

The sigma-delta (:2 b• ADC is the heart of the HF+ design and the primary reason for its exceptional dynamic range. Understanding its architecture is essential to appreciating why this receiver outperforms competing designs based on conventional successive-approximation (SAR) or flash ADCs at equivalent sample rates.

6.1 Sigma-Delta Modulation Principles

A sigma-delta modulator converts an analogue input to a high-speed, low-resolution digital bitstream — 1-bit or multi-bit — relying on massive oversampling and noise shaping to achieve high effective resolution. The critical insight is that quantisation noise is not reduced in total power, but is spectrally shaped by the modulator's feedback topology so that noise power is pushed to high frequencies (above the signal band), where it is later removed by the digital decimation filter.

The STA709 implements a 4th-order multi-bit sigma-delta modulator operating at 36 MSPS. The "4th-order" descriptor means there are four cascaded integrator stages in the loop filter, each providing one order of noise shaping. The quantisation noise power spectral density at the modulator output is proportional to (1 " z {1) t in the Z-domain, which steeply attenuates noise at low frequencies while pushing it to high frequencies near the sampling rate.

6.2 The 4th-Order Feedback Loop in Detail

The four integrator stages create a feedback structure in which the quantisation error from the output is continuously fed back and subtracted from the input, effectively predicting and cancelling the quantisation noise in an iterative process. Each additional order of the loop filter improves the noise shaping by approximately 20 dB per decade at low frequencies, with a corresponding improvement in SNDR (Signal-to-Noise-and-Distortion Ratio) within the signal band.

The multi-bit internal quantiser (as opposed to a 1-bit design) provides two additional benefits: it reduces the magnitude of each quantisation step (improving linearity of the feedback DAC), and it allows higher gain in the loop filter without risk of instability — a common challenge with high-order 1-bit sigma-delta designs.

6.3 Noise Shaping and Effective Resolution

The theoretical SNDR achievable by an Nth-order sigma-delta with a B-bit internal quantiser oversampled by ratio OSR is:

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For an Nth-order, B-bit (internal) :2 b ÖöGVÆ F÷#

SNR_max "H 6.02·B + 1.76 + 10·log • €(OSR)·(2N+1) "  10·log • €(< !Ó¢òƒ$â³ ’’ D

For N=4 (4th-order), OSR = 36e6 / 768e3 "H 47:
  Oversampling gain:  10·log • €(47) = 16.7 dB
  4th-order shaping boost: ~4×16.7 = ~67 dB total from oversampling+shaping

With hypothesised 4-bit internal quantiser (B=4, 16 levels):
  B × 6.02 = 24 dB from quantiser resolution
  Total theoretical SNR "H 24 + 67 + corrections "H 111 dB

This matches measured BDR of 110 dB and is consistent with:
  - "24-bit analogue resolution" claim (24 × 6.02 = 144 dB theoretical max)
  - Practical limit of ~110-113 dB from clock jitter, thermal noise, component tolerances
  - Output effective resolution of 18-bits = 6.02 × 18 = 108 dB
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6.4 The "24-Bit Analogue" Claim — Interpretation

AirSpy describes the sigma-delta modulator as providing "24-bit equivalent analogue resolution." This claim requires careful technical interpretation. The modulator does not produce 24 independent binary digits per sample at its native 36 MSPS rate. Rather, the combination of its noise shaping, oversampling, and the subsequent decimation chain produces an Effective Number of Bits (ENOB) equivalent to a 24-bit linear ADC operating over the full signal bandwidth.

The term "analogue resolution" refers specifically to the dynamic range of the analogue modulator loop before decimation losses. A 4th-order multi-bit sigma-delta modulator, operating at 36 MSPS with the degree of noise shaping described, can theoretically distinguish signal components with a dynamic range approaching 140 dB — equivalent to 24-bit linear resolution — provided sufficient decimation is applied. After decimation to 768 kSps, practical non-idealities (clock jitter, thermal noise, DAC non-linearity in the feedback path) reduce the effective output resolution to 18 bits (108 dB), which still far exceeds what any 16-bit linear ADC at the same final sample rate could achieve.

> **Note:**
> The crucial practical benefit of 24-bit analogue resolution is protection against ADC overload. A strong signal that would clip a 16-bit ADC (96 dB full-scale range) will still be within the sigma-delta's analogue handling range. This means strong out-of-band signals can be tolerated without saturation, giving the AGC system time to respond and adjust gain — a critical advantage in dense HF environments.

6.5 Why Sigma-Delta Over Conventional ADCs?

At 36 MSPS, a conventional 16-bit SAR ADC would provide a theoretical SNR of approximately 96 dB, limited by practical component tolerances to closer to 85–90 dB in a real implementation. A sigma-delta operating at the same clock rate with 4th-order noise shaping achieves 110+ dB within its signal band. The fundamental trade-off is reduced instantaneous bandwidth (600 kHz alias-free vs. ~18 MHz for a Nyquist SAR at the same clock), which is perfectly suited to the HF+ use case where typically one to a few MHz of spectrum is of interest at any given time.

Furthermore, the sigma-delta's inherent oversampling significantly reduces sensitivity to clock aperture jitter. For a Nyquist ADC, clock jitter of <0 seconds limits SNR to " 20·log • €(2< e÷6–væ ËsÃ). For an oversampled sigma-delta, the effectiv jitter requirement is relaxed by the oversampling ratio, making the architecture inherently more robust to clock quality — particularly important at the HF frequencies where jitter from digital switching circuitry could otherwise degrade performance.

7. Decimation Chain and IQ Output

The sigma-delta ADC produces data at 36 MSPS with most of its quantisation noise concentrated at high frequencies due to noise shaping. To extract usable narrow-band data, this raw bitstream must be low-pass filtered and decimated — the sample rate reduced by a factor that corresponds to the desired output bandwidth.

7.1 The Decimation Process

The HF+ uses a cascade of Cascaded Integrator-Comb (CIC) filters and Finite Impulse Response (FIR) filters for the decimation task. The total decimation ratio is approximately 47 (36,000,000 / 768,000 "H 47), reducing the sample rate from 36 MSPS to 768 kSps while simultaneously increasing effective resolution from ~6 bits (raw sigma-delta output) to ~18 bits (after noise shaping and averaging effects of decimation).

CIC filters are ideal as the first decimation stage because they are computationally trivial at high sample rates — implementable with only adders and delay registers, requiring no multipliers. A CIC filter provides a sin(x)/x (sinc) frequency response that naturally attenuates high-frequency energy, including the shaped quantisation noise from the sigma-delta. Each CIC stage reduces the sample rate by a factor of 2 to 32 while increasing effective resolution by approximately 0.5 bit per factor-of-2 decrease in sample rate.

FIR compensation filters follow the CIC stages to correct the CIC's inherent passband droop (the sinc roll-off) and provide additional alias suppression. The final FIR filter delivers the 108 dBc alias rejection figure published for the HF+, which is the measure of how well spurious signals outside the signal band are rejected in the output data stream.

7.2 The IQ Output: 2 × 16-Bit Architecture

The final output of the HF+ is an IQ (in-phase and quadrature) complex data stream. IQ representation separates the received signal into two orthogonal baseband components, enabling software-defined demodulation of any modulation scheme — AM, LSB, USB, FM, PSK, FT8, or any other mode — without hardware demodulators.

Although the effective resolution after decimation is 18 bits, the USB output is packaged as 2×16-bit signed integers (one for I, one for Q). The upper 16 bits of the 18-bit data word are transmitted; the two LSBs are discarded. This truncation introduces approximately 6 dB of additional quantisation noise beyond the 18-bit ADC performance, reducing effective output dynamic range to approximately 96 dB. This is equivalent to a 16-bit linear ADC at its theoretical maximum SNR — still representing excellent performance for a USB audio-class device.

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USB data rate calculation:
  768,000 samples/sec × 2 channels (I/Q) × 2 bytes/channel (16-bit)
  = 3,072,000 bytes/sec "H 3.07 MB/s

This comfortably fits within USB 2.0 High-Speed bandwidth (480 Mbit/s "H 60 MB/s max),
requiring no compression. The device enumerates as a USB audio-class device.

Alias-free bandwidth at 768 kSps:
  Theoretical Nyquist: 384 kHz each side = 768 kHz total
  Usable (alias-free): 300 kHz each side = 600 kHz total
  Transition band: 84 kHz each side (FIR filter roll-off region)
  !’ Outer 12% of displayed bandwidth may show filter roll-off effects.
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8. Overall Performance Analysis

8.1 Key Measured Performance Figures

Parameter	HF+	HF+ Discovery	Notes
MDS (500 Hz BW, 15 MHz)	" 138 dBm	" 140 dBm	Discovery benefits from integrated preselector
MDS (500 Hz BW, FM band)	" 140 dBm	" 141.5 dBm	
MDS (500 Hz BW, VHF aviation)	" 141 dBm	" 142.5 dBm	
Blocking Dynamic Range (HF)	110 dB	110 dB	Best-in-class consumer SDR at launch
Blocking Dynamic Range (VHF)	95 dB	95 dB	
IIP3 (max gain, HF)	+15 dBm	+15 dBm	Excellent for consumer device
System NF (without preselector)	~2.0 dB	—	HF+ base unit
System NF (with preselector)	~2.8 dB	~1.4 dB	Discovery integrates preselector optimally
Alias/image rejection	108 dBc	108 dBc	After CIC/FIR decimation chain
Alias-free bandwidth	600 kHz	600 kHz	At max 768 kSps output rate
Clock accuracy	0.5 ppm	0.5 ppm	Precision MEMS oscillator
Frequency coverage	9 kHz–31 MHz, 60–260 MHz	Same	HF + VHF paths

8.2 Why This Architecture Achieves Exceptional HF Performance

The HF+'s performance advantage is not attributable to any single design choice but to a coherent system-level optimisation in which each stage is matched to the capabilities of adjacent stages:

1. Low insertion-loss preselector !’ LNA NF dominates: The filter's loss (<3 dB) is sufficiently low that the excellent LNA noise figure sets the system limit. Any preselector with higher insertion loss would worsen sensitivity proportionally.
2. High LNA gain !’ subsequent stages irrelevant to NF: 20 dB of gain means all post-LNA NF contributions are divided by 100 before reaching the antenna input — rendering the VGA, mixer, and ADC noise figures negligible in the Friis cascade.
3. PHR mixer !’ no tracking filter needed: The polyphase structure provides harmonic rejection without the insertion loss and tuning complexity of a tracking bandpass filter, preserving the low-loss signal path while keeping the preselector architecture simple and consistent.
4. Low-IF architecture !’ no DC spur, no 1/f noise: The signal is never placed at DC in the analogue domain, avoiding the two most troublesome artefacts of direct-conversion designs without any additional circuit complexity beyond the IF offset in the LO.
5. 4th-order :2 b D2 " Pxtraordinary dynamic range in limited bandwidth: The ADC is precisely matched to the application: wide dynamic range (110 dB) over a narrow but sufficient bandwidth (600 kHz), which is exactly what HF monitoring requires.
6. Adaptive IF offset !’ avoids ADC clock spurs: The firmware-controlled IF shift ensures the sigma-delta's internal switching artefacts can never align with the signal of interest, maintaining a consistently clean noise floor across all tuned frequencies.
7. Multi-stage cascaded AGC !’ dynamic optimisation: Three interlocked AGC loops continually adjust gain

distribution to keep the signal in the optimal range for all stages simultaneously, without any manual operator intervention.

8.3 Comparative Context

SDR Receiver	HF BDR (approx.)	Architecture	Price Tier
RTL-SDR (R820T2)	50–55 dB	Direct sampling, 8-bit ADC	Budget
SDRplay RSPdx	~85 dB	Low-IF, 14-bit ADC	Mid-range
Microtelecom Perseus	~100 dB	Superheterodyne, 24-bit :2 `	Professional
AirSpy HF+ / Discovery	110 dB	Low-IF PHR + 4th-order :2 `	Mid-range
Top-end professional receivers	115–125 dB	Multi-conversion, precision ADC	Professional

At 110 dB, the AirSpy HF+ achieves a
performance begins to approach profe
effectiveness of the STA709-based arc

9. Potential Improvements and Their Limitations

Given the exceptional performance already achieved by the HF+ design, further improvements are necessarily incremental. Each potential improvement carries trade-offs that must be weighed against the primary use case requirements.

9.1 Preselector Filter Enhancement

Potential Improvement:
More aggressive bandpass filters (7- or 9-pole LC designs, or ceramic resonator filters) could improve adjacent-band rejection by 15–25 dB. However, each additional filter pole adds approximately 0.1–0.3 dB insertion loss, which directly degrades system noise figure since the preselector precedes the LNA. A 0.5 dB increase in preselector loss would raise system NF from 1.4 to 1.9 dB — a measurable sensitivity penalty. For operation in extremely congested environments (e.g., near MW/AM broadcast transmitters with >100 kW ERP), this trade-off may be worth accepting.

9.2 External LNA Preamplifier

Adding an external LNA preamplifier before the HF+ can reduce the effective system noise figure further. Independent measurements show approximately 0.5–1 dB of improvement with a quality external preamp (e.g., a low-NF MMIC with 15–20 dB gain). However, this improvement is at the margin of practical utility on HF:

Note:
Below approximately 10 MHz, the HF noise floor is dominated by external noise sources: atmospheric noise, man-made interference, and galactic background radiation. At these frequencies, even the best receiver cannot improve upon what the antenna delivers. The HF+ already achieves a noise figure of ~1.4 dB, corresponding to a noise temperature of ~120 K. At HF, external noise temperatures routinely exceed 10,000 K. Improving receiver NF below ~2 dB at HF provides no measurable practical benefit — the antenna is the limiting factor.

9.3 Wider Instantaneous Bandwidth

Potential Improvement:
The 600 kHz alias-free bandwidth limits the spectrum observable at any instant. Increasing this to 2–4 MHz would require a higher-order decimation chain or a higher USB transfer rate. The STA709 can output data at rates above 768 kSps, and a USB 3.0 interface would handle the increased throughput (a 4 MHz IQ stream at 16-bit depth requires ~128 Mbit/s, well within USB 3.0 capability). This would be the single most useful architectural improvement for wideband monitoring applications such as spectrum analysis or signal hunting across multiple bands simultaneously.

9.4 Reference Clock Improvement

The 0.5 ppm clock accuracy is excellent for most applications, including SSB reception where carrier offset within 100 Hz is typically acceptable. For precision frequency measurement, frequency standard comparison, or meteor scatter work requiring exact Doppler measurement, an external GPSDO (GPS-Disciplined Oscillator) reference reducing uncertainty to 0.001 ppm or better would be beneficial. Most software defined radio applications already support external 10 MHz reference clock injection — this is an external improvement available without hardware modification.

9.5 Summary Assessment

The HF+ and HF+ Discovery are already highly optimised designs. The engineering trade-offs are well-reasoned: the active VGA adds negligible noise post-LNA; the PHR mixer eliminates tracking filter complexity; the low-IF architecture cleanly avoids DC artefacts; and the sigma-delta ADC provides extraordinary dynamic range matched to the use case bandwidth. Any improvements are refinements to an already excellent system, not fundamental corrections to architectural choices.

10. Conclusion

The AirSpy HF+ and HF+ Discovery represent a coherent, systems-level engineering achievement in consumer SDR design. By selecting and combining an automotive-grade sigma-delta ADC IC (STMicroelectronics STA709) with a carefully designed preselector filter bank, a high-gain low-noise LNA (MiniCircuits PSA4-5043+), a sophisticated multi-stage AGC system, and a low-IF architecture with polyphase harmonic rejection mixing, the AirSpy design team achieved a blocking dynamic range of 110 dB on HF — unprecedented in the consumer SDR market at launch and remaining best-in-class for devices in this price tier.

The paper has examined each signal chain stage in detail. The dominant noise contribution is the preselector filter's insertion loss (1.4 dB), not the active stages; the LNA gain ensures subsequent stages contribute negligibly to system NF. The PHR mixer eliminates the historical necessity for a frequency-tracking front-end filter, maintaining low insertion loss while providing adequate harmonic rejection across the entire HF band. The low-IF architecture avoids the DC spur and 1/f noise problems endemic to zero-IF designs. The 4th-order multi-bit sigma-delta ADC provides 24-bit analogue resolution, protecting against ADC overload while delivering 18-bit effective resolution after decimation. The three-loop AGC architecture dynamically optimises gain distribution without any manual operator adjustment.

Potential improvements — steeper preselector filters, external LNA preamplifiers, wider bandwidth, better reference clock — are real but minor in the context of the primary application. The filter NF trade-off limits preselector improvements; the HF noise floor renders further LNA NF improvements impractical; wider bandwidth requires a higher-speed interface but is architecturally feasible; clock accuracy is already excellent for most use cases.

The HF+ stands as a compelling demonstration that consumer-grade SDR performance can approach professional receiver standards through thoughtful exploitation of automotive-grade silicon, rigorous cascade noise analysis, and careful system-level co-design between the analogue front end, the ADC, and the digital signal processing chain. It achieves in a pocket-sized USB dongle what required rack-mounted professional equipment a decade earlier.

Note:
All performance figures in this paper are based on published manufacturer specifications and independent third-party measurements by radio amateurs and technical reviewers. Internal design details not publicly disclosed (filter topology, specific IF value, AGC algorithm details, internal quantiser bit depth) are estimated through engineering analysis consistent with published performance data, and are explicitly labelled as hypotheses or engineering inferences throughout the paper.

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